

12/8/26

Activity -

S.A. Mohammed
Faisel

19261288

b)

Given input { 4371, 1323, 6173, 4199, 4344, 9679, 1989 }
and a hash function $h(x) = x \pmod{10}$ show result

a) Separate chaining hash table

Key	$x \pmod{10}$
4271	1
1323	3
6173	3
4199	9
4344	4
9679	9
1989	9

Hash table:

index	chain
0	-
1	4271
2	-
3	1323 → 6173
4	4344
5	-
6	-
7	-
8	-
9	4199 → 9679 → 1989

b) Open addressing - Linear Probing

$$\text{Formula} : h(x) = h(x) + i \mod 10$$

$$1) 4271 \% 10 = 1$$

$$2) 1323 \% 10 = 3$$

$$3) 6173 \% 10 = 3 \Rightarrow i=0 \rightarrow (3+0) \mod 10$$

$$\Rightarrow i=1, (3+1) \mod 10 = 4$$

$$6173 \rightarrow 4$$

$$4) 4199 \% 10 = 9$$

$$5) 4344 \% 10 = 4 \Rightarrow i=1, (4+1) \mod 10 = 5$$

$$4344 \rightarrow 5$$

$$6) 9679 \% 10 = 9 \Rightarrow (9+1) \mod 10 = 0$$

$$9679 \rightarrow 0$$

$$7) 1989 \% 10 = 9 \Rightarrow (9+3) \mod 10 = 2$$

$$1989 \rightarrow 2$$

Final Table:

Index	Value
0	9679
1	4271
2	1989
3	1323
4	6173
5	4344
6	-
7	-
8	-
9	4199

Open addressing - Quadratic Probing

$$\text{Formula} = h(x) = (h(x) + i^2) \bmod 10$$

1) 4271 $\% 10 = 1$

2) 1323 $\% 10 = 3$

3) 6173 $\% 10 = 3 \Rightarrow (3 + 1^2) \bmod 10 = 4$

6173 $\Rightarrow 4$

4) 4199 $\% 10 = 9$

5) 4344 $\% 10 = 4 \Rightarrow (4 + 1^2) \bmod 10 = 5$

4344 $\Rightarrow 5$

6) 9679 $\% 10 = 9 \Rightarrow (9 + 1^2) \bmod 10 = 0$

9679 $\Rightarrow 0$

7) 1989 $\% 10 = 9 \Rightarrow (9 + 2^2) \bmod 10 = 8$

1989 $\Rightarrow 8$

Final Table:

Index Value

0 9679

1 4271

2 -

3 1323

4 6173

5 4344

6 -

7 -

8 1989

9 4199



d) Double Hashing

$$h_1(\text{key}) = \text{key} \bmod 10$$

$$h_2(\text{key}) = h_1(\text{key}) + I \cdot [R - (h_1(\text{key}) \bmod R)] \bmod 10$$

$R=7$

1) $4271 \cdot 10 = 1$

2) $1323 \cdot 10 = 3$

3) $6173 \cdot 10 = 3 \rightarrow (3+1 \cdot 1) \bmod 10 = 4$
 $6173 \rightarrow 4$
 $R - (6173 \bmod 7) \Rightarrow 7 - 6 = 1$

4) $4199 \cdot 10 = 9$
 ~~$R - (4199 \bmod 7) = 7 - 6 = 1$~~
 ~~$(9+1 \cdot 1) \bmod 10$~~

5) $4344 \cdot 10 = 3$
 $R - (4344 \bmod 7) \Rightarrow 7 - 4 = 3$

$I=0, (4+0 \cdot 3) \bmod 10 = 4$
 Index 4 has 6173

$I=1, (4+1 \cdot 3) \bmod 10 = 7$
 $4344 \rightarrow 7$

6) $9679 \cdot 10 = 9$
 $R - (9679 \bmod 7) = 7 - 5 = 2$
 $I=3, (9+3 \cdot 2) \bmod 10 = 5$
 $9679 \rightarrow 5$

7) $1989 \cdot 10 = 9$
 $R - (1989 \bmod 7) = 7 - 1 = 6$

$I=5, (9+5 \cdot 6) \bmod 10 = 9 \rightarrow$ sequence repeats
 1989 cannot be inserted

Final Table:

Index	Value
0	-
1	4271
2	-
3	1323
4	6173
5	9679
6	-
7	4344
8	-
9	4199

Amir
 12/12/20